Class Overview

Meiying Zhang

COE201 - AI Large Models

- Lectures: Monday 10:20-12:10AM
- Software labs: Monday 4:20-6:10PM
- Office hours: Thursday 2:30-3:30PM
- For personal /sensitive matters: zhangmy@sustech.edu.cn

Who am I?

1. Artificial Intelligence and Deep Learning

2. Trustworthy Autonomous Driving Systems

3.Design of robotic bionics and control systems

Program Staff

Instructor：张美莹

Co-Instructor：彭维源

TA：宋天佑

TA：曹始文

Learning Goals

What is the goal of this course ?

It's not just about learning how to use Large Models to do your homework!!!

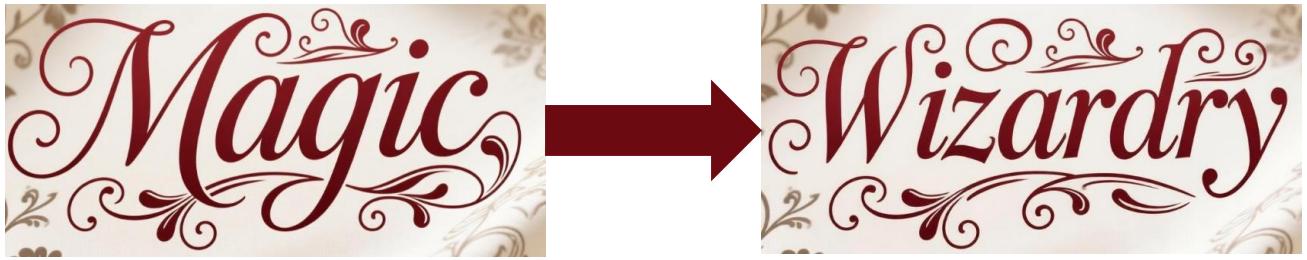
How to use these large model tools.

User → Interpreter → Creator

Learning Goals

What is the benefits of knowing the principles?

Large models are incredibly powerful, like performing magic—everyone can create amazing results.



Once you understand the principles, magic is no longer mystical; you can see through the cleverness and even create your own tricks.

Learning Goals

After successfully completing this course you will be able to:

1. Understand the basics behind large models as a “magical black box," such as transformer architecture and the self-attention mechanism.
2. Explain the common terms used in academic papers on LLMs (e.g., alignment, scaling laws, RLHF, prompt engineering, instruction tuning).
3. Compare and contrast different models in the LLM ecosystem in order to determine the best model for any given task.
4. Utilize open-source libraries to fine-tune and do inference with popular pre-trained language models.
5. Apply LLMs to a variety of downstream applications.

Course Planning

	Contents	Hours
Chapter1: Overview of AI and Large Models	1.1 History of AI 1.2 Fundamentals of Deep Learning 1.3 Evolution of Sequence Modeling (RNN/LSTM→ Seq2Seq → Transformer)	6
Chapter2: Principles and Architectures of Large Models	2.1 Transformer Components 2.2 From Transformer to Large Language Models: Encoder-Only, Encoder-Decoder, Decoder-Only 2.3 Training and Inference Processes	6
Chapter3: Prompting, Reasoning, and Evaluation of Large Models	3.1 Fundamentals and Patterns of Prompt Engineering: In-Context Learning 3.2 Reasoning Optimization Techniques: Quantization, Knowledge Distillation, Pruning, etc. 3.3 Performance Evaluation of Large Models	6

Course Planning

	Contents	Hours
Chapter4: Pre-training and Fine-Tuning of Large Models	4.1 Pre-training: Data Collection, Cleaning, Distributed Training, etc. 4.2 Aligning Models with Human Intent: Core Concepts of Instruction Fine-Tuning, and Parameter-Efficient Fine-Tuning (PEFT)	6
Chapter5: Applications of Large Models	5.1 API Invocation: Function Calling / Tools Calling 5.2 Retrieval-Augmented Generation (RAG) 5.3 Multimodal Large Models: ViT, CLIP, etc.	6

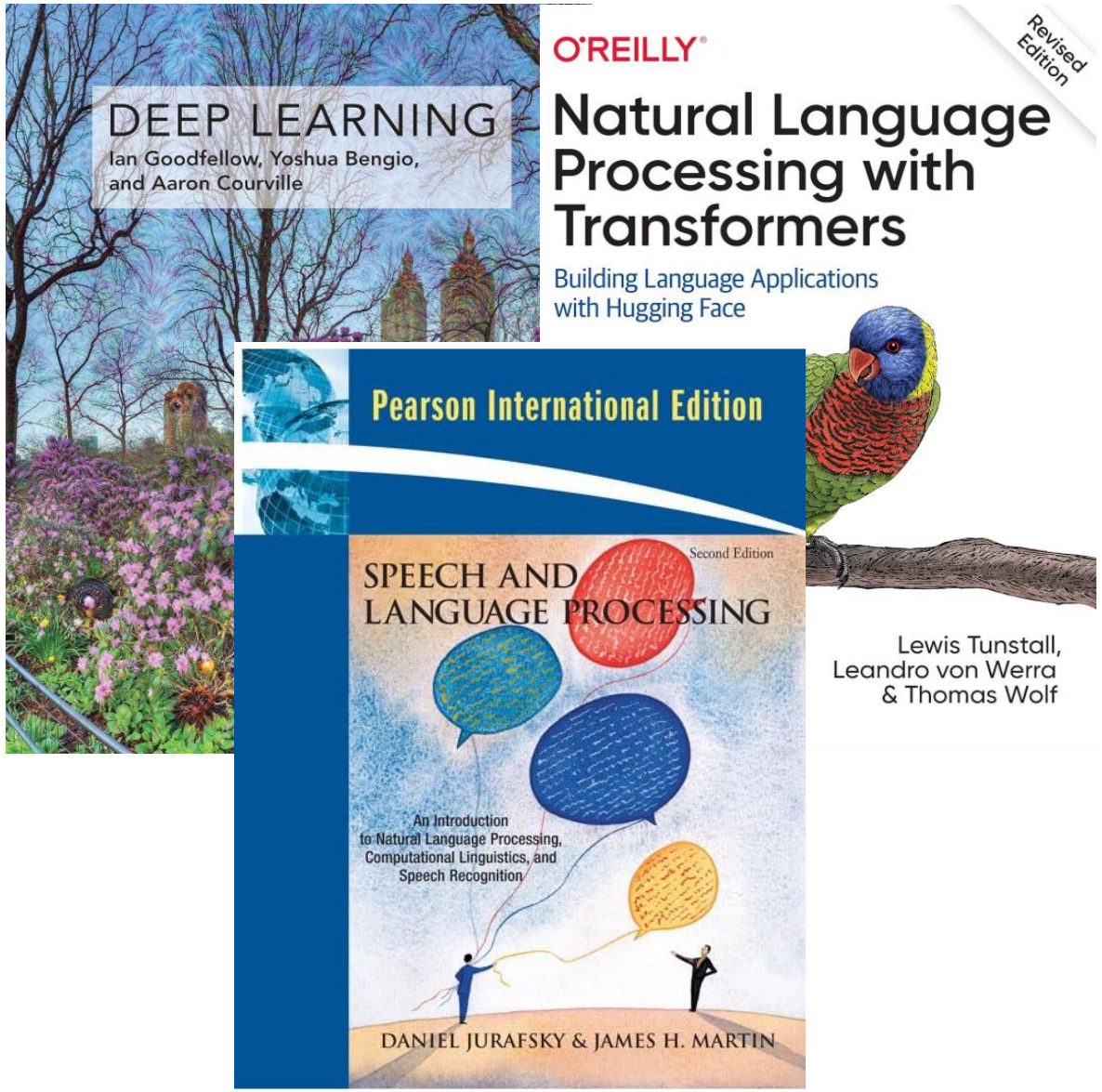
Assessment

- 5% Attendance (5 times)
- 15% Quiz (3-5 items in class)
- 25% Projects (2-3 people/group, presented in class during one of the lectures)
- 20% Assignments (TBD after class)
- 35% Final Exam (written)

Assessment

- Each student submits assignment in next class
- These allow to take extra time due to unforeseen circumstances (e.g., illnesses).
- In more serious cases, prioritize your safety, health, and wellbeing. (Talk to us!)
- A homework submitted outside of late days applied → 0 points

Readings



What Language Model Architecture and Pretraining Objective Work Best for Zero-Shot Generalization?

Thomas Wang[1*] Adam Roberts[2*] Daniel Hesslow[3] Teven Le Scao[1] Hyung Won Chung[2] Iz Beltagy[4]

Abstract

Large pretrained Transformer languag
have been shown to exhibit *zero-shot*
ization, i.e. they can perform a wid
of tasks that they were not explicitl
on. However, the architectures and
ing objectives used across state-of-the
els differ significantly, and there has
ited systematic comparison of these fa
this work, we present a large-scale e
of modeling choices and their impact
shot generalization. In particular, we
text-to-text models and experiment w
model architectures (causal/non-causal
only and encoder-decoder), trained wit
ferent pretraining objectives (autoregre
masked language modeling), and evalu
and without multitask prompted finetu
train models with over 5 billion paran
more than 168 billion tokens, thereby i
the likelihood that our conclusions wil
to even larger scales. Our experiments
causal decoder-only models trained on
gressive language modeling objective e
strongest zero-shot generalization aft
self-supervised pretraining. However
with non-causal visibility on their inp
with a masked language modeling obje
lowed by multitask finetuning perform
among our experiments. We therefore
the adaptation of pretrained models acr
tectures and objectives. Code and ch
are available at https://github.com/bi
workshop/architecture-objective.

*Equal contribution †Equal supervisio
Face [2]Google [3]LightOn [4]Allen Institute fo
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Learning, Baltimore, Maryland, USA, PMLR 16
right 2022 by the author(s).

What Are Tools Anyway? A Survey from the Language Model Perspective

Zora Zhiruo Wang◆ Zhoujun Cheng◆ Hao Zhu◆ Daniel Fried◆ Graham Neubig◆
◆Carnegie Mellon University ◆Shanghai Jiao Tong University

Language models (LM
Tools have substantial
complex skills. Howe
ways, raising the ques
how do tools help LMs?
tools as external progr
of LM tooling scenari
empirically study the
their required compu
and highlight some ch

1 Introduction

Language Models (LMs) have
by taking in natural language
such as answering the "What i
LMs often struggle to perform
soning), and are fundamentally
not included in their training

To solve this problem, researc
ers are turning to LMs enhanc
help *facilitate* the task-solving
extend LMs with new abilities
possess otherwise (Qin et al.,
2023). For example, a `calcu`
used to facilitate mathematic
`get_time()` tool could be use
rent time, which is not availa
the LM's parameters. Inspire
by humans (Shumaker et al.,
introduce application-specific
such as using a `search engi`
edge (Lazaridou et al., 2022; K
languages (Schick et al., 2023),
Zhuang et al., 2023). With the
faces (APIs) on the web, many
(Balog et al., 2016; Xu et al.,
2023), even performing profe
and digital marketing (Huang
implementations, other works
(Wang et al., 2024a; Cao et al.,
can be created by human (Gu
Cai et al., 2023; Wang et al., 20

However, despite this broad a
cover certain tool categories s

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis[†‡], Ethan Perez[*],
Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],
Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel[†‡], Sebastian Riedel[†‡], Douwe Kiela[†]

Scaling Laws for Neural Language Models

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Abstract

We study empirical scaling laws for language model performance on the cross-entropy loss. The loss scales as a power-law with model size, dataset size, and the amount of compute used for training, with some trends spanning more than seven orders of magnitude. Other architectural details such as network width or depth have minimal effects within a wide range. Simple equations govern the dependence of overfitting on model/dataset size and the dependence of training speed on model size. These relationships allow us to determine the optimal allocation of a fixed compute budget. Larger models are significantly more sample-efficient, such that optimally compute-efficient training involves training very large models on a relatively modest amount of data and stopping significantly before convergence.

Contact

该二维码在7天内(3月6日前)有效
重新进入将更新